FR306X 显存单双 buffer 应用说明

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FREQCHIP 富芮坤

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1. 概述

刷屏在 RGB 屏幕驱动过程占用了大部分的主频带宽，为了释放更多的主频带宽给系统，启动 DMA 托管 RGB 刷屏。用户可根据 UI 的实际复杂度选择显存单 buffer 或双 buffer 框架使用。

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2. 双 buffer 模式

2.1 RGB 驱动回环逻辑设计，为了确保 Vsync 的前肩后肩一致性，使用 DMA 的中断处理方式，保证了帧与帧之间的数据更新完整性。

1.DMA init & DMA IRQ enable。

```
__DISPLAY_RGB_SET_PIXEL_CLK_DIV(hdisplay.DISPLAYx,2);
printf("display div clock:%d\r\n", system_get_peripheral_clock(PER_CLK_DISPLAY));
/* DMA Init */
system_dmac_request_id_config(DISPLAY_TRANS,DMA0_REQUEST_ID_2);
dma_display_handle.DMAx                          = DMA0;
dma_display_handle.Channel                       = DMA_Channel1;
dma_display_handle.Init.Data_Flow                = DMA_M2P_DMAC;
dma_display_handle.Init.Request_ID               = 2;
dma_display_handle.Init.Source_Master_Sel        = DMA_AHB_MASTER_2;
dma_display_handle.Init.Desination_Master_Sel    = DMA_AHB_MASTER_3;
dma_display_handle.Init.Source_Inc               = DMA_ADDR_INC_INC;
dma_display_handle.Init.Desination_Inc           = DMA_ADDR_INC_NO_CHANGE;
dma_display_handle.Init.Source_Width             = DMA_TRANSFER_WIDTH_32;
dma_display_handle.Init.Desination_Width         = DMA_TRANSFER_WIDTH_16;
dma_display_handle.Init.Source_Burst_Len         = DMA_BURST_LEN_32;
dma_display_handle.Init.Desination_Burst_Len     = DMA_BURST_LEN_32;
dma_init(&dma_display_handle);


NVIC_ClearPendingIRQ(DMA0_IRQn);
NVIC_EnableIRQ(DMA0_IRQn);
```

2.DMA start。

```
__RAM_CODE void rgb_controller_reinit(void)
{
    rgb_display_controller_init();
}

void st7282_init(void* buffer)
{
    rgb_display_controller_init();      tang zheng, 4个月前 • create new project …
    rgb_lcd_backlight_set();
    __DISPLAY_RGB_ENABLE(hdisplay.DISPLAYx);
    dma_start_IT(&dma_display_handle, (uint32_t)buffer & 0xf7ffffff,
        (uint32_t)&hdisplay.DISPLAYx->TX_FIFO, PIXEL_SIZE/4);

}
```

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3.DMA irq process,RGB 驱动是使用 psram 双 buffer 方式，在 GUI 渲染完成后需 SWAP buffer 驱动屏幕。

```
__RAM_CODE void st7282_rgb_display_dma_irq(void)
{
    static uint32_t cnt = 0;
    if (dma_get_tfr_Status(&dma_display_handle))
    {
        dma_clear_tfr_Status(&dma_display_handle);
        te_sign = 1;
        disp_dma_finish = 1;        tang zheng, 4个月前 • create new project
//      if( !refr_wait_sign)
        {
            // printf("d %d",frame_buffer_index);
            if(frame_buffer_index){

                //hrgb->rgb_TxData = get_display_buffer2();
                dma_start_IT(&dma_display_handle, (uint32_t)get_display_buffer2() & 0xf7ffffff, (uint32_t)&hdisplay.DISPLAY
            }else{
                //hrgb->rgb_TxData = get_display_buffer1();
                dma_start_IT(&dma_display_handle, (uint32_t)get_display_buffer1() & 0xf7ffffff, (uint32_t)&hdisplay.DISPLAY
            }
        }

    }
}
```

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2.2 full buffer 渲染和区块渲染的逻辑设计，方便客户提升 GUI 渲染的帧率做优化处理，侧重客户对 GUI 的需求实际应用情况。

1.refr_en 状态是为 true 时，每一次渲染都同时更新双 bufrer 的内容，相反则只更新当前渲染 buffer 的内容。用户可以根据实际 UI 页面都 refr_en 控制，达到优化提升帧率的目的。

```
#ifdef REFLASH_PSRAM_BUFFER
refr_en = false;
#endif
for(i = 0; i < disp_refr->inv_p; i++) {
    /*Refresh the unjoined areas*/
    if(disp_refr->inv_area_joined[i] == 0) {

        if(i == last_i) disp_refr->driver->draw_buf->last_area = 1;
        disp_refr->driver->draw_buf->last_part = 0;

        #ifdef REFLASH_PSRAM_BUFFER
        if((disp_refr->inv_areas[i].x2 +1 - disp_refr->inv_areas[i].x1 == 800)
            && (disp_refr->inv_areas[i].y2 +1 - disp_refr->inv_areas[i].y1 == 480))      You, 1秒钟前 • Uncommitted changes
            refr_en = true;
        #endif

        refr_area(&disp_refr->inv_areas[i]);

        px_num += lv_area_get_size(&disp_refr->inv_areas[i]);
    }
}


frame_buffer_index ^= 1;

#ifdef REFLASH_PSRAM_BUFFER
if(refr_en)
{

    vTaskDelay(1);

    for(i = 0; i < disp_refr->inv_p; i++) {
        /*Refresh the unjoined areas*/
        if(disp_refr->inv_area_joined[i] == 0) {

            if(i == last_i) disp_refr->driver->draw_buf->last_area = 1;
            disp_refr->driver->draw_buf->last_part = 0;

            refr_area(&disp_refr->inv_areas[i]);

            px_num += lv_area_get_size(&disp_refr->inv_areas[i]);
        }
    }
}
```

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2.disp_drv.full_refresh 可配置满 buffer 渲染还是区块渲染，可根据 UI 的实际情况提升帧率。

```
lv_init();

lv_disp_draw_buf_init(&disp_buf, (void *)display_framebuffer, NULL, LV_HOR_RES_MAX * DRAW_BUFFER_LINE);    /*Initialize the d
//lv_disp_draw_buf_init(&disp_buf, (void *)display_framebuffer1, (void *)display_framebuffer2, LV_HOR_RES_MAX * LV_VER_RES_MA

/* Implement and register a function which can copy the rendered image to an area of your display */
static lv_disp_drv_t disp_drv;          /*Descriptor of a display driver*/
lv_disp_drv_init(&disp_drv);            /*Basic initialization*/
disp_drv.flush_cb = my_disp_flush;      /*Set your driver function*/
disp_drv.draw_buf = &disp_buf;          /*Assign the buffer to the display*/
disp_drv.hor_res = LV_HOR_RES_MAX;
disp_drv.ver_res = LV_VER_RES_MAX;
disp_drv.physical_hor_res = -1;
disp_drv.physical_ver_res = -1;      You, 上个月 • 1.更新sdk
disp_drv.offset_x = 0;
disp_drv.offset_y = 0;
disp_drv.full_refresh = 1;
lv_disp_drv_register(&disp_drv);        /*Finally register the driver*/
/* Implement and register a function which can read an input device. E.g. for a touch pad */

static lv_indev_drv_t indev_drv;        /*Descriptor of a input device driver*/
lv_indev_drv_init(&indev_drv);          /*Basic initialization*/
indev_drv.type = LV_INDEV_TYPE_POINTER; /*Touch pad is a pointer-like device*/
indev_drv.read_cb = my_touchpad_read;   /*Set your driver function*/
lv_indev_drv_register(&indev_drv);      /*Finally register the driver*/
```

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联系方式

欢迎大家针对富芮坤产品和文档提出建议。

反馈：doc@freqchip.com

网站：www.freqchip.com

销售：sales@freqchip.com

电话：+86-21-5027-0080

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